

Infron 与 OpenRouter 路由、缓存、成本与流式性能 A/B 实验报告

摘要

本报告评估 `openai/gpt-5.4-mini` 在 Infron 与 OpenRouter 的 Prompt Caching 工作负载表现。实验使用 `/v1/chat/completions`，保留模型与平台默认 `reasoning/thinking` 行为，并启用 `prompt` 长度分层。

过滤后保留 792 个配对样本 和 3168 条请求级观测，剔除 16 条记录。

核心结论：Infron 在吞吐量上全部路由模式占优，并在多数路由模式下成本更低；OpenRouter 在端到端 E2E 时延和 Streaming TTFT 上整体更低。缓存命中率两边均处于高位，不同路由模式下优势方有所变化。

质量门禁

项目	值
模型	<code>openai/gpt-5.4-mini</code>
API 协议	仅 <code>/v1/chat/completions</code>
组数 / 轮数	4 组 x 50 轮
Prompt 分层	<code>short</code> about 1500, <code>medium</code> about 8000, <code>long</code> about 32000 tokens
Reasoning / thinking	平台与模型默认行为；请求中不显式禁用
配对规则	strict sort/group/round pair, first/second <code>usage.prompt_tokens</code> 差异均不超过 50
保留配对样本	792
请求级观测	3168
剔除记录	16

路由级结果矩阵

路由模式	配对样本	Token 命中率	实际成本	吞吐量	E2E 时延	TTFT
吞吐优先	200	Infron 96.68% / OpenRouter 97.91%	Infron \$0.665196 / OpenRouter \$0.736837	Infron 17.822 tok/s / OpenRouter 5.076 tok/s	Infron 4042.33 ms / OpenRouter 2737.70 ms	Infron 2722.39 ms / OpenRouter 2337.97 ms
价格优先	200	Infron 97.99% / OpenRouter 97.95%	Infron \$0.549838 / OpenRouter \$0.643934	Infron 18.259 tok/s / OpenRouter 4.922 tok/s	Infron 3896.09 ms / OpenRouter 2830.58 ms	Infron 2763.05 ms / OpenRouter 2492.73 ms

路由模式	配对样本	Token 命中率	实际成本	吞吐量	E2E 时延	TTFT
端到端时延优先	192	Infron 97.98% / OpenRouter 96.73%	Infron \$0.514386 / OpenRouter \$0.642328	Infron 20.873 tok/s / OpenRouter 5.189 tok/s	Infron 3465.18 ms / OpenRouter 2690.54 ms	Infron 2473.68 ms / OpenRouter 2350.08 ms
流式 TTFT 优先	200	Infron 96.43% / OpenRouter 97.99%	Infron \$0.868611 / OpenRouter \$0.579755	Infron 25.861 tok/s / OpenRouter 6.077 tok/s	Infron 2694.75 ms / OpenRouter 2301.99 ms	Infron 2141.42 ms / OpenRouter 2098.17 ms

API 协议记录

API 协议	Endpoint	说明
chat_completions	/v1/chat/completions	本轮仅保留该标准 Chat Completions 协议；不包含 /v1/messages 或 /v1/responses。

Prompt 长度分层缓存表现

分层	配对样本	Token 命中率	调用命中率	E2E 时延	TTFT	Reasoning tokens
short		Infron 0.00% / OpenRouter 0.00%	Infron 0.00% / OpenRouter 0.00%	Infron 0.00 ms / OpenRouter 0.00 ms	Infron 0.00 ms / OpenRouter 0.00 ms	Infron 0 / OpenRouter 0
medium		Infron 0.00% / OpenRouter 0.00%	Infron 0.00% / OpenRouter 0.00%	Infron 0.00 ms / OpenRouter 0.00 ms	Infron 0.00 ms / OpenRouter 0.00 ms	Infron 0 / OpenRouter 0
long		Infron 0.00% / OpenRouter 0.00%	Infron 0.00% / OpenRouter 0.00%	Infron 0.00 ms / OpenRouter 0.00 ms	Infron 0.00 ms / OpenRouter 0.00 ms	Infron 0 / OpenRouter 0

Reasoning / Thinking 观测

本轮不显式禁用或强制 reasoning/thinking；reasoning tokens 按响应 telemetry 观测。

路由模式	Reasoning tokens	Avg reasoning tokens/request
吞吐优先	Infron 0 / OpenRouter 0	Infron 0.0000 / OpenRouter 0.0000
价格优先	Infron 0 / OpenRouter 0	Infron 0.0000 / OpenRouter 0.0000
端到端时延优先	Infron 0 / OpenRouter 0	Infron 0.0000 / OpenRouter 0.0000
流式 TTFT 优先	Infron 0 / OpenRouter 0	Infron 0.0000 / OpenRouter 0.0000

可复现性引用

工件	GitHub 路径	SHA256 / 说明
Summary	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/openai/gpt-5.4-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/data/summary.json	b5fcdd6787abbc331d2a4326ce6473a0bd40226faa9adc3bf370bfa1c3
配对数据集 CSV	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/openai/gpt-5.4-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/data/benchmark_pairs.csv	3bd26737e7ca08438856d3d9f6367df63ac391adc7e0db841c81ae34a3
请求级数据集 JSONL	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/openai/gpt-5.4-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/data/benchmark_requests.jsonl	4ff92a2e44c3d37cf6eb732ca00a85393114c4712ec35780b884899c55
剔除记录审计	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/openai/gpt-5.4-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/data/records_excluded.json	11e2ed24815a7cf38765b2149b9d28bd51f312a36dbcedad6ef6d32b16
Benchmark 执行源码	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/openai/gpt-5.4-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/code/rerun_routing_sort_cache_cost_ab.py	a485c5116e99375e6b171d0328e5a423babd1ee6b206c72f54053a3a04
HTML 报告渲染源码	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/openai/gpt-5.4-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/code/render_glm52_deepseek_style_report.py	e5bb0d32361d70ee688e0d0a0da3302a1e542e33e53876c905814eb5c2

工件	GitHub 路径	SHA256 / 说明
测试源码	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/openai/gpt-5.4-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/code/test_rerun_routing_sort_cache_cost_ab.py	c086ddc5d0a9a91eba82b7e8767d7bddf2f1ca4a28af0a87056b702882
Reports 目录	https://github.com/InfronAI/prompt-cache-bench/tree/main/experiments/openai/gpt-5.4-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/reports	中英文 HTML / Markdown / PDF 报告
Data 目录	https://github.com/InfronAI/prompt-cache-bench/tree/main/experiments/openai/gpt-5.4-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/data	配对数据、请求级 telemetry、summary 与剔除样本
Manifest	https://github.com/InfronAI/prompt-cache-bench/blob/main/experiments/openai/gpt-5.4-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/metadata/manifest.json	文件大小与 SHA-256 checksum

数据集引用： `business_representative` 内置代表性业务模板；请求级导出见 `benchmark_requests.jsonl`。

在线 HTML：中文 https://infronai.github.io/prompt-cache-bench/experiments/openai/gpt-5.4-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/reports/routing-cache-cost-streaming-performance-ab-study_gpt-5-4-mini_infron-vs-openrouter_4x50-stream-ttft-reasoning-default-length-tiers-chat-completions_2026-07-04.zh.html；英文 https://infronai.github.io/prompt-cache-bench/experiments/openai/gpt-5.4-mini/infron-vs-openrouter-routing-sort-cache-cost-4x50-stream-ttft-reasoning-default-length-tiers-chat-completions-2026-07-04/reports/routing-cache-cost-streaming-performance-ab-study_gpt-5-4-mini_infron-vs-openrouter_4x50-stream-ttft-reasoning-default-length-tiers-chat-completions_2026-07-04.en.html。